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# When Adapter Training Hurts: Manifold Preservation for Vector Retrieval

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## Abstract

Adapting pre-trained vision-language models for vector retrieval typically relies on global contrastive objectives, which optimize semantic alignment across all training categories simultaneously. We identify a fundamental limitation of this approach: when applied through lightweight adapters, global objectives produce dense gradient fields that systematically distort the latent manifold of unseen categories, causing out-of-distribution (OOD) retrieval performance to collapse below the frozen baseline. Although parameter-efficient low-rank adapters such as LoRA largely preserve performance when trained with global losses like InfoNCE or triplet objectives due to their capacity bottleneck, they do not exploit local sparse updates to the same extent. To address this, we propose Euclidean Geodesic Alignment (EGA), a residual adapter trained with a small-margin triplet loss on the class-aware neighborhood graph of pre-trained embeddings. By operating locally rather than globally, EGA converges to a sparse-gradient state in which 96.5% of training triplets produce zero gradient at steady state, leaving the broader embedding space, including regions occupied by unseen categories, structurally intact. Extensive evaluations on CIFAR-10, CIFAR-100, FGVC-Aircraft, and Food-101 demonstrate that EGA consistently outperforms global contrastive methods including ICon and SRL across both in-distribution and OOD settings while also surpassing or matching the LoRA+InfoNCE and LoRA+Triplet baselines. On strictly unseen categories, EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10 and bounds the degradation on Food-101, while ICon and SRL fall below the frozen baseline by up to 36%.

## 1 Introduction

Modern retrieval systems increasingly rely on embedding models such as CLIP [1] for their remarkable zero-shot semantic capabilities. These models provide highly generalizable latent spaces that serve as excellent starting points for vector similarity search. However, when deploying these systems to specialized domains, practitioners often seek to further align the embeddings with domain-specific category structures to maximize retrieval accuracy. Because full scale fine-tuning is computationally prohibitive and risks destroying the foundational zero-shot knowledge, parameter-efficient adaptation has emerged as the standard paradigm. In this approach, lightweight adapters are trained on a labeled subset of the target task while the massive pre-trained backbone remains frozen.

Despite the elegance of this parameter-efficient paradigm, the training of these adapters is almost universally driven by global contrastive objectives. These methods compute losses over the full pairwise similarity matrix in each mini batch, continuously reshaping the entire embedding distribution to enforce global alignment and uniformity. While these objectives successfully boost retrieval accuracy for in-distribution queries where all relevant categories are present during training, they introduce a critical failure mode for real-world retrieval systems that must inevitably handle unseen

categories. We reveal that when paired with high-capacity adapters, global contrastive methods such as ICon [2] and SRL [3] systematically distort the pre-trained semantic manifold of unseen classes. Although the resulting embeddings appear geometrically optimized for IVF-based indexing, they frequently retrieve neighbors of the wrong semantic class. This distortion causes out-of-distribution Label Precision to drop by 4% to 36% below the raw frozen baseline across multiple benchmarks. In contrast, while low-rank adapters such as LoRA [4] largely avoid this catastrophic OOD degradation, their restricted capacity inherently limits their ability to fully capture complex semantic alignments on seen domains. Consequently, practitioners face a fundamental dilemma: scaling adapter capacity under global objectives trades semantic correctness for geometric indexability, whereas restricting capacity compromises overall retrieval efficacy.

To resolve this fundamental dilemma, we propose Euclidean Geodesic Alignment (EGA), a manifold-preserving adapter that breaks the tradeoff between model capacity and semantic integrity by replacing dense global optimization with sparse local updates. EGA incorporates three synergistic design choices: (i) a zero-initialized residual structure, (ii) an  $\ell_2$  normalization constraint, and (iii) a small-margin triplet loss operating on a class-aware neighborhood graph. Together, these components restrict the adapter’s influence to local neighborhood relationships that explicitly violate the target retrieval objective. As training proceeds, the fraction of triplets producing a non-zero gradient decays rapidly. At convergence, 96.5% of the training triplets contribute zero gradient. Consequently, the high-capacity adapter ceases to modify the vast majority of the embedding space, successfully preserving the pre-trained semantic structure for unseen categories and achieving bounded degradation even under heavy out-of-distribution shifts.

We evaluate EGA across in-distribution (CIFAR-100) and three out-of-distribution benchmarks (CIFAR-10, FGVC-Aircraft, Food-101). We compare EGA against the frozen CLIP baseline, global contrastive adapters (ICoN and SRL), and low-rank adapters (LoRA). On in-distribution data, EGA consistently improves retrieval quality. On unseen categories, EGA improves Label Precision over the frozen baseline on Aircraft and CIFAR-10 and bounds degradation on Food-101. In contrast, ICoN and SRL fall below the frozen baseline on every out-of-distribution dataset, dropping by 32% to 36% on Food-101. While LoRA avoids this collapse, it underperforms EGA due to capacity constraints. On FGVC-Aircraft, EGA achieves a Label Precision of 0.548, outperforming the frozen baseline (0.512), ICoN (0.423), and SRL (0.441).

**Contributions** This paper makes the following contributions: (i) We identify that dense optimization gradients from global contrastive adapters distort the pre-trained manifold of unseen categories; (ii) We propose Euclidean Geodesic Alignment (EGA) to refine metric geometry via a zero-initialized residual adapter and local triplet loss; and (iii) We demonstrate that EGA consistently improves out-of-distribution Label Precision over the frozen baseline, outperforming state-of-the-art methods by a margin of up to 36%.

## 2 Related Work

**Manifold Preservation under Adapter Tuning** Foundation models such as CLIP [1] produce embedding spaces whose structure encodes both seen class semantics and a broader prior over visual concepts not observed during downstream training. Catastrophic forgetting has been studied extensively in continual and incremental learning settings [5, 6]. Recent work has documented analogous degradation phenomena specific to contrastive representations: anisotropic collapse into narrow conic regions [7, 8], structural fracturing under modality gap [9, 10], and loss of compositional structure under aggressive optimization [11, 12, 13]. To counter these degradations, recent adaptation strategies impose explicit geometric constraints such as orthogonality regularization [14] or restrict optimization to limited subspaces. EGA leverages the inherent sparsity of triplet loss optimization to confine adapter updates to local neighborhoods of seen classes, leaving the unseen class regions of the pretrained manifold structurally unmodified.

**Global Contrastive Objectives and Structural Regularization** Recent advancements in representation learning often rely on global contrastive objectives to optimize the latent space [15, 16]. A common paradigm involves jointly enforcing alignment for positive pairs and uniformity for the overall feature distribution [17], a strategy applied across various domains including graph encoders [18] and multimodal recommendation systems [19]. While pushing embeddings toward a uniform distribu-

tion enhances robustness to background noise [20], it inherently assumes that all latent regions require the same degree of global scattering. To further refine complex distributions, structural regularization techniques introduce auxiliary constraints. For instance, methods like ICon formulate a unifying framework to systematically align multi view representations [2], while geometric constraints are leveraged to improve imbalanced regression tasks [3]. Other approaches attempt to decouple the learning objectives [21] or apply information theoretic regularizers [22] to control the density of the learned representations. Furthermore, region based distillation techniques have been proposed to enhance fine grained understanding [23]. EGA replaces dense global optimization with sparse local triplet updates, preserving out of distribution semantic structure.

**Parameter-Efficient Manifold Adaptation** Parameter efficient adaptation emerged as a standard to prevent catastrophic forgetting and reduce computational overhead when transferring large foundation models [24, 4]. This paradigm has extensively shifted towards vision language models, dominating recent literature due to the prohibitive cost of full scale fine tuning [25]. Methods such as Visual Prompt Tuning [26] and Tip Adapter demonstrate that lightweight tuning can effectively specialize representations for downstream tasks without altering the frozen backbone [27]. Furthermore, extensive benchmarking confirms that parameter efficient techniques can rival full fine tuning while preserving the generalizable representations of the foundation model [28]. To further protect the pre trained manifold during initial training steps, zero initialization strategies have proven highly effective. For instance, conditional control mechanisms in image generation initialize adapter weights to zero, ensuring the model behaves exactly like the frozen base model before any optimization occurs [29]. EGA integrates a zero initialized residual adapter with a localized geometric objective, achieving both parameter efficiency and targeted manifold smoothing.

**Vector Similarity Search and Metric Consistency** Vector similarity search is the foundational engine for retrieving high dimensional representations at scale. The systems community has developed highly optimized indexing algorithms to handle billion scale datasets. Foundational works such as FAISS leverage hardware acceleration for fast similarity search [30], and Hierarchical Navigable Small World graphs establish the core topology for efficient approximate retrieval [31]. Meanwhile, hybrid structures like DiskANN and SPANN achieve exceptional speed by efficiently managing memory and disk hierarchies [32, 33]. Recent advancements continue to refine these topologies, exploring crossing sparse proximity graphs [34] and analyzing the theoretical constructions of navigable graphs for high dimensional nearest neighbor search [35]. Furthermore, system level studies optimize retrieval latency and stability by aligning best first search algorithms with solid state drives [36] and enabling direct insertions in large scale indices [37]. To reduce the computational cost of exact distance calculations during the retrieval phase, approximation techniques such as low rank matrix factorization have also been introduced to accelerate inner product evaluations [38]. However, extensive theoretical analyses reveal that popular approximate nearest neighbor implementations still face severe worst case performance limitations [39]. EGA proactively recalibrates the metric manifold prior to index construction, enabling standard search backends to achieve superior semantic recall without requiring complex algorithmic reengineering.

## 3 Methodology

### 3.1 Problem Formulation

Let  $\phi : \mathcal{X} \rightarrow \mathbb{R}^d$  denote a frozen vision-language encoder (e.g., CLIP ViT-B/32) that maps an image  $x \in \mathcal{X}$  to a unit-normalized embedding  $\mathbf{z} = \phi(x) \in \mathbb{S}^{d-1}$ . We are given a set of labeled training images  $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^N$ , where  $y_i \in \mathcal{Y}_{\text{seen}}$  belongs to a set of *seen* classes. At inference time, the retrieval system must operate over a database that also contains images from a disjoint set of *unseen* classes  $\mathcal{Y}_{\text{unseen}}$ , with  $\mathcal{Y}_{\text{seen}} \cap \mathcal{Y}_{\text{unseen}} = \emptyset$ .

Our goal is to learn a lightweight adapter  $f_\theta : \mathbb{S}^{d-1} \rightarrow \mathbb{S}^{d-1}$  such that the transformed embeddings  $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$  support more accurate approximate nearest neighbor (ANN) retrieval on *both* seen and unseen classes, without access to any unseen-class labels during training.

**Evaluation metrics.** We evaluate retrieval quality along two complementary dimensions.

Label Precision@ $K$  measures the semantic quality of retrieved results—whether the  $K$  nearest neighbors in the transformed space share the same class label as the query:

$$\text{LP@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{1}{K} \sum_{k=1}^K \mathbf{1}[y_{\hat{n}_k(q)} = y_q], \quad (1)$$

where  $\mathcal{Q}$  is the query set and  $\hat{n}_k(q)$  denotes the  $k$ -th nearest neighbor of  $q$  in the transformed embedding space.

ANNS Recall@ $K$  measures the geometric fidelity of the ANN index—the fraction of true  $K$  nearest neighbors (under exact search) that are recovered by the approximate index at a given search budget  $n_{\text{probe}}$ :

$$\text{AR@}K = \frac{1}{|\mathcal{Q}|} \sum_{q \in \mathcal{Q}} \frac{|\mathcal{N}_K^{\text{exact}}(q) \cap \mathcal{N}_K^{\text{approx}}(q)|}{K}, \quad (2)$$

where  $\mathcal{N}_K^{\text{exact}}(q)$  and  $\mathcal{N}_K^{\text{approx}}(q)$  denote the exact and approximate top- $K$  neighbor sets, respectively.

The two metrics capture distinct failure modes: an adapter may improve LP@ $K$  by tightening within-class clusters (semantic improvement) while simultaneously improving AR@ $K$  by making the geometry more amenable to IVF-based indexing (geometric improvement). As we show in Section 4, global contrastive objectives achieve high AR@ $K$  on seen classes but collapse on *both* metrics for unseen classes, whereas EGA improves both simultaneously under the OOD setting.

### 3.2 Manifold-Preserving Adapter Design

EGA introduces a lightweight adapter  $f_\theta$  composed of stacked residual blocks with a final refinement layer. The architecture is deliberately constrained by three design principles, each of which serves a distinct role in preserving the pretrained manifold.

**Residual connection with zero initialization.** Each EGA block applies a residual update of the form

$$\mathbf{z}' = \mathbf{z} + g_\theta(\mathbf{z}), \quad (3)$$

where  $g_\theta$  is a learned transformation initialized so that  $g_\theta(\mathbf{z}) \approx \mathbf{0}$  at the start of training. Concretely, the final linear layer of  $g_\theta$  is initialized to zero, making  $f_\theta$  an identity mapping at initialization. This guarantees that training begins from the pretrained geometry rather than a random perturbation, and that the adapter can only *refine* rather than replace the CLIP embedding. As confirmed by our ablation (Table 2), removing the residual connection causes accuracy to collapse from 0.70 to 0.01, because the lightweight MLP cannot reconstruct complex semantic relationships from seen-class data alone.

**Feature gating.** Inside each residual block, the transformed signal passes through a feature-gating module:

$$\text{Gate}(\mathbf{h}) = \mathbf{h} \odot \sigma(W_2 \text{GELU}(W_1 \mathbf{h})), \quad (4)$$

where  $W_1 \in \mathbb{R}^{(d/4) \times d}$ ,  $W_2 \in \mathbb{R}^{d \times (d/4)}$ ,  $\sigma$  is the sigmoid function, and  $\odot$  denotes elementwise multiplication. This structure, analogous to Squeeze-and-Excitation networks [40], learns a per-dimension importance weight that amplifies retrieval-relevant dimensions and suppresses noisy ones. The bottleneck ratio of 1/4 is chosen to prevent the gating network from overfitting on seen-class data, as validated in Table 3.

**$\ell_2$  normalization.** The final output of  $f_\theta$  is projected back onto the unit hypersphere  $\mathbb{S}^{d-1}$  via  $\ell_2$  normalization:

$$\hat{\mathbf{z}} = \frac{f_\theta(\mathbf{z})}{\|f_\theta(\mathbf{z})\|_2}. \quad (5)$$

This constraint ensures that the transformed embeddings remain in the same metric space as the original CLIP embeddings, so that Euclidean distance and cosine similarity remain interchangeable on the hypersphere. Removing this constraint degrades retrieval performance by 4.2 percentage points (Table 2), confirming that hypersphere consistency is necessary for stable ANN indexing.

179 **Full forward pass.** Let  $\text{Block}_i$  denote the  $i$ -th EGA block consisting of a two-layer MLP with LayerNorm followed by feature gating, and let  $\text{Refine}$  denote the final linear projection with LayerNorm. 180 The complete forward pass for an input embedding  $\mathbf{z} \in \mathbb{S}^{d-1}$  is: 181

$$\hat{\mathbf{z}} = \ell_2(\text{Refine}(\text{Block}_2(\text{Block}_1(\mathbf{z})))) . \quad (6)$$

182 The total number of trainable parameters is approximately  $4.2M$  for  $d = 512$  and hidden dimension 183 2048, which is negligible compared to the frozen CLIP backbone ( $\sim 150M$  parameters).

### 184 3.3 Local Triplet Training and Gradient Sparsity

185 **Training objective.** Given a mini-batch sampled from  $\mathcal{D}_{\text{train}}$ , we construct triplets  $(a, p, n)$  where  $a$  186 is an anchor image,  $p$  is a positive sampled uniformly from the same class as  $a$ , and  $n$  is a negative 187 sampled uniformly from a different class. The adapter is trained with the standard triplet margin loss:

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \max(0, d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m), \quad (7)$$

188 where  $d(\cdot, \cdot)$  denotes Euclidean distance,  $m > 0$  is the margin hyperparameter, and  $\hat{\mathbf{z}} = f_\theta(\mathbf{z})$  is the 189 transformed embedding. We set  $m = 0.2$  throughout all experiments based on the sensitivity analysis 190 in Table 3.

191 **Gradient sparsity and manifold preservation.** A triplet  $(a, p, n)$  contributes a non-zero gradient 192 to  $\theta$  if and only if the margin constraint is *violated*:

$$d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n) + m > 0. \quad (8)$$

193 We refer to such triplets as *active*. As the adapter converges on seen classes, increasingly many 194 triplets satisfy the margin constraint and become inactive, contributing zero gradient. Figure 5 (left) 195 shows that the active triplet ratio decays from 17.4% at epoch 10 to 3.5% at convergence. At steady 196 state, **96.5% of triplets produce zero gradient**, meaning the adapter ceases to modify the vast 197 majority of local neighborhood relationships in the embedding space.

198 This sparsity has a direct consequence for out-of-distribution generalization. Because unseen classes 199 are absent from  $\mathcal{D}_{\text{train}}$ , no triplet involving an unseen-class embedding is ever constructed. Provided 200 that the adapter’s updates remain local (i.e., that gradients from seen-class triplets do not propagate 201 into unseen-class regions of the embedding space), the CLIP manifold structure for unseen classes 202 is preserved intact. The gradient sparsity mechanism enforces exactly this locality: once seen-class 203 neighborhoods satisfy the margin, the corresponding gradients vanish, and only the small fraction of 204 actively-violated local relationships continue to be updated.

205 **Contrast with global contrastive objectives.** Global objectives such as ICon [2] and SRL [3] 206 compute losses over the *full* pairwise similarity matrix within each mini-batch. ICon minimizes a KL 207 divergence between the empirical similarity distribution and the class-membership distribution, while 208 SRL jointly optimizes batch-wide uniformity and within-class homogeneity. Both objectives generate 209 dense gradient fields in every training step: every sample pair (regardless of whether its local geometry 210 already satisfies the retrieval objective) contributes to the loss and receives a non-zero gradient. This 211 global optimization pressure from seen-class training propagates throughout the embedding space, 212 distorting the local manifold structure that CLIP had established for unseen categories.

213 **Proposition 1** (Gradient Sparsity at Convergence). *Let  $f_\theta$  be trained with the triplet loss in Eq. (7) 214 with margin  $m > 0$  on a dataset  $\mathcal{D}_{\text{train}}$  drawn from seen classes  $\mathcal{Y}_{\text{seen}}$ . Define the active triplet ratio 215 at training step  $t$  as*

$$\rho(t) = \Pr_{(a,p,n) \sim \mathcal{B}} \left[ d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_p^{(t)}) - d(\hat{\mathbf{z}}_a^{(t)}, \hat{\mathbf{z}}_n^{(t)}) + m > 0 \right]. \quad (9)$$

216 *As  $f_\theta$  converges,  $\rho(t) \rightarrow \rho^*$ , where*

$$\rho^* = \Pr_{(a,p,n)} \left[ d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m > 0 \right] \quad (10)$$

217 *is determined solely by the margin  $m$  and the within-class distance distribution of the converged 218 embeddings. Furthermore,  $\nabla_\theta \mathcal{L} = \mathbf{0}$  for all triplets  $(a, p, n)$  with  $d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_p^*) - d(\hat{\mathbf{z}}_a^*, \hat{\mathbf{z}}_n^*) + m \leq 0$ , so 219 the adapter gradient is supported only on the fraction  $\rho^*$  of the training distribution. For embeddings 220 whose local geometry already satisfies the margin constraint, the adapter exerts zero update pressure.*

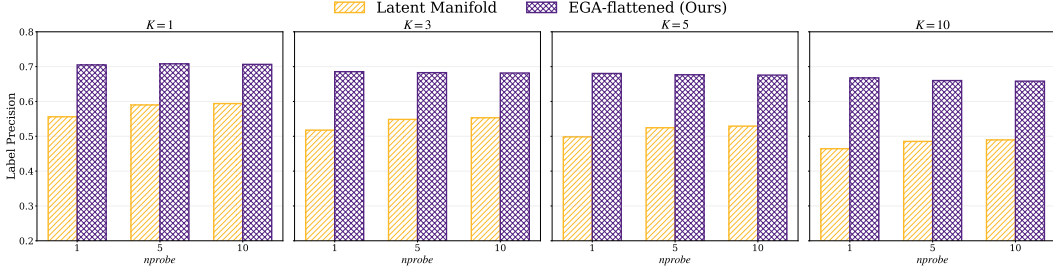


Figure 1: Label Precision@ $K$  on CIFAR-100 across  $K \in \{1, 3, 5, 10\}$  and  $nprobe \in \{1, 5, 10\}$ . EGA consistently improves over the raw CLIP latent space.

*Proof Sketch.* The triplet loss consists of hinge terms whose gradient is zero whenever the margin constraint is satisfied. At convergence, only those triplets whose geometry the adapter has not yet separated within the margin remain active. The fraction of such triplets is determined solely by the margin  $m$  and the distance distribution of seen classes, independent of unseen classes. Since no training triplet involves unseen-class embeddings, the adapter exerts zero gradient over the entire unseen-class region of the embedding space. See Appendix A for the complete proof.  $\square$

## 4 Evaluation

We answer the following research questions through extensive experiments:

- RQ1: Does EGA improve retrieval quality for in-distribution queries, and how does it affect the geometric structure of the embedding space? (Section 4.2)
- RQ2: Under strict out-of-distribution settings, how does EGA perform on both Label Precision and ANNS Recall compared to global contrastive methods and low-rank adapters? (Section 4.3)
- RQ3: What is the underlying mechanism that enables triplet trained adapters to preserve unseen class structures while global contrastive objectives destroy them? (Section 4.4)
- RQ4: How do the individual design choices and hyperparameters impact the overall retrieval effectiveness and stability of EGA? (Section 4.5)

### 4.1 Experimental Setup

Experiments were conducted on the Chameleon Cloud platform using a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA A100 GPU of 80 GB HBM2e. Evaluation datasets include CIFAR-100 for in distribution scenarios, CIFAR-10 for cross dataset OOD settings, and FGVC-Aircraft and Food-101 for unseen class OOD evaluations. All of these datasets can be directly downloaded from the Pytorch library. We compare our method against four baseline models: ICon [2], SRL [3], LoRA + InfoNCE [4, 41], and LoRA + Triplet [4].

Extended experimental details can be found in Appendix B.

### 4.2 In-Distribution Validation

**Semantic retrieval quality.** Figure 1 reports Label Precision@ $K$  across four neighborhood sizes and three search budgets. EGA improves Label Precision@1 from  $\approx 0.55$  (raw CLIP) to  $> 0.70$  at  $nprobe = 1$ , and the gap persists across all  $K$  and  $nprobe$  values. The improvement is not an artifact of larger search budgets: even at  $nprobe = 10$ , EGA retains a clear advantage, indicating that the adapter genuinely tightens within-class neighborhoods rather than merely benefiting from broader index coverage.

**Geometric structure.** The Label Precision improvement has a direct geometric correlate. Figure 2 contrasts the  $L_2$  distance distributions for true Top- $K$  neighbors against random background points, before and after EGA. In the raw CLIP space, the two distributions overlap substantially: the distance

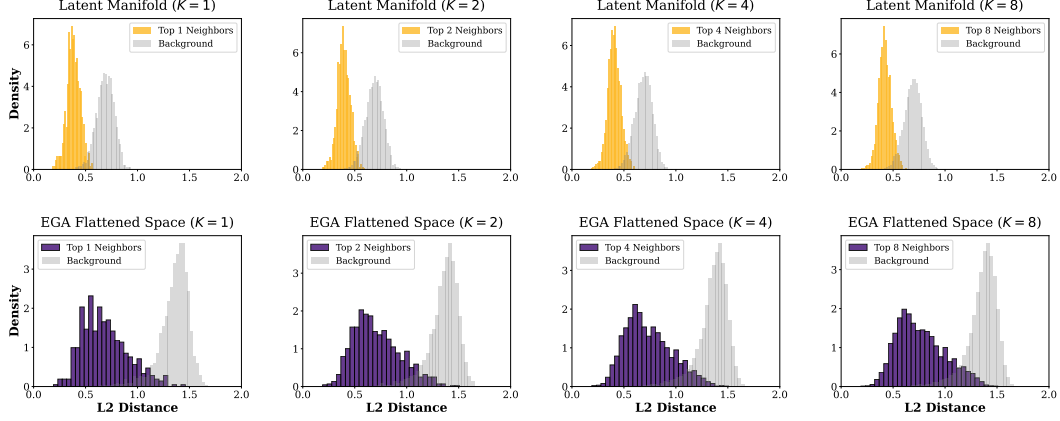


Figure 2:  $L_2$  distance distributions for Top- $K$  neighbors versus random background points on CIFAR-100. The raw CLIP space exhibits significant overlap; EGA separates the two distributions cleanly.

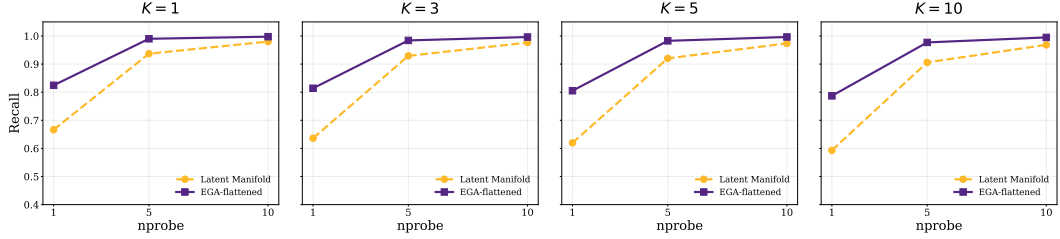


Figure 3: ANNS Recall@ $K$  versus nprobe on CIFAR-100. EGA shifts the efficiency frontier upward across all  $K$ .

to a true neighbor is often indistinguishable from the distance to an unrelated point, which is exactly the condition under which IVF-based ANN indexing degrades. After EGA, the two distributions separate cleanly across all  $K$ , with neighbors concentrated at small distances and background points pushed toward a distinct mode. This separation is the mechanism by which EGA improves the recall–efficiency frontier shown in Figure 3: the raw CLIP space achieves only 0.67 Recall@1 at nprobe = 1, whereas EGA reaches 0.80 under identical indexing, and saturates above 0.97 by nprobe = 5.

### 4.3 Out-of-Distribution Generalization

We now turn to the central empirical claim of this paper. Under strict OOD settings where evaluation categories are disjoint from training categories, the two retrieval metrics tell different stories about adapter quality, and EGA is the only method that performs well on both. Table 1 reports headline numbers at  $K = 1$ , nprobe = 1, and Figure 4 traces Label Precision across nprobe  $\in \{1, 5, 10\}$  to show that the pattern is robust to search budget. We evaluate on three benchmarks: CIFAR-10 (trained on CIFAR-100), FGVC-Aircraft (80 seen / 20 unseen classes), and Food-101 (80 seen / 21 unseen classes). For a fair comparison, all methods train identical lightweight adapters on top of frozen CLIP features and differ only in their loss functions.

On ANNS Recall, every adapter improves or matches the frozen CLIP baseline across all three OOD benchmarks (Table 1). Adapter training succeeds at *geometric* reorganization: the resulting embedding distribution is more amenable to IVF-based indexing. On Label Precision, however, the picture inverts. ICon and SRL fall *below* the frozen CLIP baseline on every dataset, with degradation reaching  $-32$  to  $-36$  percentage points on Food-101. The retrieved neighbors are geometrically closer in the IVF index, but they belong to the wrong semantic class. EGA is the only adapter that improves Label Precision on Aircraft (+3.6%) and CIFAR-10 (+0.6%), and on Food-101 it retains the strongest Label Precision among all adapters with bounded  $-8.3\%$  degradation.

Table 1: OOD performance at  $K = 1$ ,  $nprobe = 1$ .

| Dataset  | Metric | CLIP  | ICon         | SRL          | LoRA+InfoNCE | LoRA+Triplet | EGA          |
|----------|--------|-------|--------------|--------------|--------------|--------------|--------------|
| CIFAR-10 | AR@1   | 0.863 | <b>0.913</b> | <b>0.942</b> | 0.856        | <b>0.866</b> | <b>0.986</b> |
|          | LP@1   | 0.880 | 0.841        | 0.794        | 0.880        | 0.870        | <b>0.886</b> |
| Aircraft | AR@1   | 0.773 | <b>0.875</b> | <b>0.863</b> | <b>0.911</b> | <b>0.881</b> | <b>0.898</b> |
|          | LP@1   | 0.512 | 0.423        | 0.441        | <b>0.560</b> | <b>0.577</b> | <b>0.548</b> |
| Food-101 | AR@1   | 0.842 | 0.811        | 0.827        | <b>0.902</b> | <b>0.899</b> | <b>0.886</b> |
|          | LP@1   | 0.881 | 0.557        | 0.520        | 0.878        | 0.833        | 0.798        |

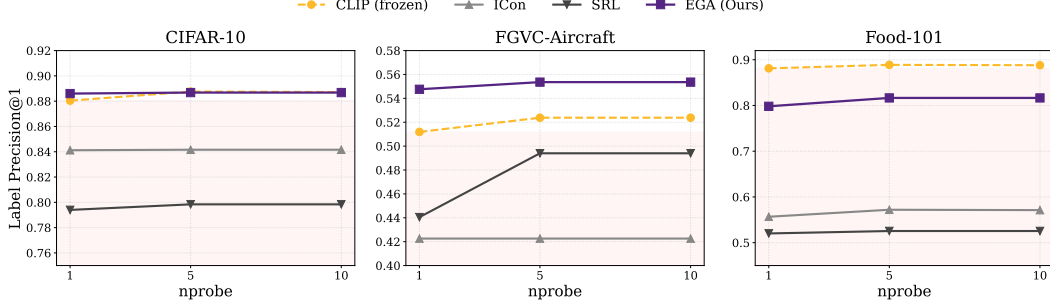


Figure 4: Label Precision@1 across three OOD benchmarks and three search budgets.

280 The Label Precision degradation of global contrastive methods persists across all three datasets and all  
 281 three search budgets we tested (18% in total, Figure 4), ruling out the possibility that it is an artifact  
 282 of a particular benchmark or indexing configuration. EGA’s relative advantage over global methods  
 283 ranges from 4.5% on CIFAR-10 to 24% on Food-101, and tracks the difficulty of the OOD shift:  
 284 when seen and unseen classes are visually similar (as in Food-101’s fine-grained dishes), the dense  
 285 gradient updates of ICon and SRL pull unseen samples most aggressively into the wrong seen-class  
 286 clusters.

287 We additionally evaluate LoRA [4] as a parameter-efficient baseline (rank  $r = 8$ ,  $\sim 8K$  trainable  
 288 parameters versus  $\sim 4.2M$  for the EGAMLP architecture used by EGA, ICon, and SRL). Two  
 289 configurations are included: LoRA + InfoNCE (a global contrastive loss) and LoRA + Triplet (the  
 290 same loss as EGA). Both LoRA configurations avoid the catastrophic Label Precision collapse  
 291 exhibited by ICon and SRL, even when paired with a global loss (LoRA + InfoNCE on Food-101  
 292 reaches LP 0.878), only 0.3% below the frozen baseline. This suggests that the catastrophic failure of  
 293 ICon and SRL arises from the *combination* of (i) full-capacity adapter and (ii) dense global gradients,  
 294 not from the global loss alone. With an aggressive bottleneck on adapter capacity, global losses  
 295 become survivable.

#### 296 4.4 Gradient Sparsity as OOD Protection

297 A triplet loss with margin  $m$  produces a non-zero gradient for a sample triplet  $(a, p, n)$  only when the  
 298 margin constraint is violated, i.e., when

$$d(a, p) - d(a, n) + m > 0. \quad (11)$$

299 As the adapter converges on seen classes, the fraction of such *active* triplets decays rapidly. Figure 5  
 300 (left) tracks this ratio throughout training: starting at 17.4% at epoch 10, it falls to 3.5% by conver-  
 301 gence. At steady state, 96.5% of triplets contribute zero gradient, meaning the adapter updates only a  
 302 sparse subset of local neighborhood relationships and leaves the remainder of the embedding space  
 303 structurally intact.

304 Global contrastive objectives such as ICon [2] and SRL [3] exhibit the opposite behavior. ICon  
 305 minimizes a KL divergence over the full pairwise similarity matrix, and SRL jointly optimizes  
 306 batch-wide uniformity and homogeneity. Both produce *dense* gradient fields in every training step:  
 307 every sample pair contributes to the loss regardless of whether its local geometry already satisfies the  
 308 retrieval objective. This global optimization pressure shapes the entire embedding distribution based  
 309 on seen-class similarities, even for regions of the space occupied by unseen-class samples. The result



Table 2: Ablation study of EGA

| Variant                 | Accuracy |
|-------------------------|----------|
| Full EGA (Ours)         | 0.7015   |
| w/o Residual connection | 0.0065   |
| w/o Zero-initialization | 0.6840   |
| w/o L2-normalization    | 0.6590   |

Table 3: Sensitivity of different hyperparameters

| Triplet Margin $m$   | 0.1    | 0.2    | 0.3    | 0.5    |
|----------------------|--------|--------|--------|--------|
| Recall@1             | 0.8420 | 0.8180 | 0.8160 | 0.7400 |
| Recall@5             | 0.9782 | 0.9654 | 0.9610 | 0.9125 |
| Hidden Dimension $d$ | 512    | 1024   | 2048   | 4096   |
| Recall@1             | 0.8300 | 0.8340 | 0.8200 | 0.8240 |
| Recall@5             | 0.9695 | 0.9712 | 0.9680 | 0.9688 |

is not loss of geometric structure but rather loss of *semantically correct* structure: unseen samples are pulled toward the clusters of seen classes that they happen to resemble in CLIP’s original metric, rather than toward other unseen samples that share their true category.

Figure 5 (right) and Figure 4 together quantify the consequence: across all three OOD benchmarks, ICon and SRL fall below the frozen CLIP baseline on Label Precision (e.g., on Aircraft, ICon collapses to 0.423 from a baseline of 0.512), confirming that dense gradient updates actively harm zero-shot semantic retrieval. EGA, by contrast, improves on Aircraft and CIFAR-10 while modifying only 3.5% of the embedding geometry at convergence.

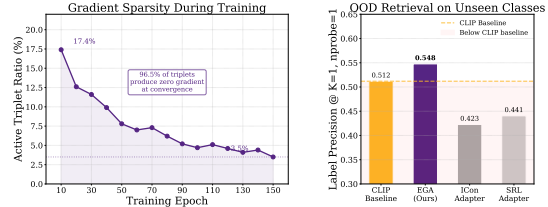


Figure 5: Gradient sparsity as OOD protection.

#### 4.5 Ablation and Sensitivity Analysis

Table 2 isolates the contribution of each architectural component. The residual connection is the core element. Removing it causes accuracy to collapse from 0.7015 to 0.0065. This collapse indicates that the adapter cannot reconstruct semantic relationships from target data alone and must operate as a refinement over the pre-trained features. The L2 normalization ensures the transformed embeddings remain on the unit hypersphere. Removing this constraint decreases accuracy to 0.6590. Zero-initialization allows the adapter to begin training as an identity mapping. Removing it lowers accuracy to 0.6840.

Table 3 evaluates model stability across hyperparameter variations. EGA maintains consistent retrieval quality across margin values from 0.1 to 0.3. Increasing the margin to 0.5 degrades Recall@1 to 0.7400. A larger margin forces the adapter to update embeddings that already satisfy local neighborhood constraints, which reduces gradient sparsity and alters the pre-trained manifold. Furthermore, the model shows stable performance across hidden dimensions ranging from 512 to 4096. This stability confirms that EGA achieves metric correction through its structural design rather than specific parameter capacities.

## 5 Conclusion

High-capacity adapters trained with global contrastive objectives harm out-of-distribution retrieval performance. Under these objectives, Label Precision on unseen categories falls below the frozen baseline. EGA resolves this failure mode by converging to a sparse gradient regime. At steady state, 96.5% of training triplets contribute zero gradient. This sparsity preserves the pre-trained semantic structure for unseen categories. Across three out-of-distribution benchmarks (CIFAR-10, FGVC-Aircraft, Food-101), EGA improves Label Precision over the frozen baseline and bounds degradation on Food-101. In contrast, ICon and SRL degrade by up to 36%. These results establish gradient sparsity as a design principle for retrieval adapters operating under distribution shift.

**Limitations.** This study focuses on visual representations from CLIP. Future work will extend EGA to text and multimodal models, explore unsupervised variants, and disentangle the interaction between adapter capacity and loss sparsity.

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## A Proof of Proposition 1

*Proof.* We prove each claim in turn.

**(1) Gradient support.** The triplet loss in Eq. (7) can be written as  $\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{(a,p,n) \in \mathcal{B}} \ell_{apn}$ , where  $\ell_{apn} = \max(0, \delta_{apn} + m)$  and  $\delta_{apn} = d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p) - d(\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_n)$ . The subgradient with respect to  $\theta$  is

$$\nabla_{\theta} \ell_{apn} = \mathbf{1}[\delta_{apn} + m > 0] \cdot \nabla_{\theta} \delta_{apn}, \quad (12)$$

which is exactly zero whenever  $\delta_{apn} + m \leq 0$ . Therefore,  $\nabla_{\theta} \mathcal{L}$  receives contributions only from active triplets satisfying Eq. (8).

**(2) Convergence of  $\rho(t)$ .** Under mild regularity conditions (Lipschitz continuity of  $f_{\theta}$  and bounded embedding norms), the sequence  $\{\hat{\mathbf{z}}^{(t)}\}$  converges to a stationary point  $\hat{\mathbf{z}}^*$  of  $\mathcal{L}$ . At stationarity,  $\rho(t)$  converges to  $\rho^*$  as defined in Eq. (10), since  $\rho(t)$  is a continuous functional of the embedding distribution and the embeddings converge.

**(3) Zero gradient over unseen regions.** Let  $\mathcal{R}_{\text{unseen}} \subset \mathbb{S}^{d-1}$  denote the region of the embedding space occupied by unseen-class images under  $f_{\theta}$ . By construction, all triplets  $(a, p, n)$  in  $\mathcal{D}_{\text{train}}$  satisfy  $y_a, y_p, y_n \in \mathcal{Y}_{\text{seen}}$ , so  $\hat{\mathbf{z}}_a, \hat{\mathbf{z}}_p, \hat{\mathbf{z}}_n \notin \mathcal{R}_{\text{unseen}}$  for all training triplets. The gradient  $\nabla_{\theta} \delta_{apn}$  depends on  $f_{\theta}$  only through its evaluation at  $\mathbf{z}_a, \mathbf{z}_p, \mathbf{z}_n$ , which lie outside  $\mathcal{R}_{\text{unseen}}$ . Consequently, no training step produces a gradient that directly modifies  $f_{\theta}$  in  $\mathcal{R}_{\text{unseen}}$ , and the CLIP geometry in that region is preserved throughout training.

**(4) Dependence on  $m$ .** The residual distance gap  $\delta_{apn}^*$  at convergence is bounded by the within-class diameter  $\Delta_c = \max_{x, x' \in c} d(\hat{\mathbf{z}}, \hat{\mathbf{z}}')$  for seen class  $c$ . For  $m \leq \min_c \Delta_c$ , the adapter can drive all within-class distances below the threshold, pushing  $\rho^*$  toward zero. For  $m > \min_c \Delta_c$ , some triplets remain active at convergence because the within-class spread exceeds the margin, resulting in a strictly positive  $\rho^*$ . This explains the empirically observed increase in  $\rho^*$  with larger  $m$  reported in Table 3.  $\square$

## B More Experimental Details

**Datasets.** We evaluate on four datasets covering both in-distribution and OOD retrieval scenarios. **CIFAR-100** (100 classes, 60K images) serves as the in-distribution benchmark: the adapter is trained and evaluated on the same class label space, with a 75/25 split between database and query sets. **CIFAR-10** (10 classes, 60K images) provides a cross-dataset OOD setting: the adapter is trained on CIFAR-100 and evaluated on the disjoint CIFAR-10 classes. **FGVC-Aircraft** (100 fine-grained aircraft variants, 10K images) and **Food-101** (101 food categories, 25K images for the test split we use) provide unseen-class OOD settings: classes are randomly partitioned into 80% seen / 20% unseen with a fixed seed, the adapter is trained on the seen split, and retrieval is evaluated strictly on the unseen split. Image embeddings for all datasets are extracted with frozen CLIP ViT-B/32.

**Baselines.** We compare EGA against the frozen CLIP baseline and four trained adapters, all sharing the same training data, evaluation protocol, and 75/25 retrieval split:

- **Icon** [2]: a global contrastive objective minimizing the KL divergence between the empirical similarity distribution and the class-membership distribution. We instantiate Icon on the same EGAMLP architecture used by our method ( $\sim 4.2$ M parameters).
- **SRL** [3]: a structural regularization objective jointly optimizing batch-wide uniformity and within-class homogeneity, also instantiated on the EGAMLP architecture.
- **LoRA + InfoNCE** [4, 41]: a low-rank adapter (rank  $r = 8$ ,  $\alpha = 16$ ,  $\sim 8$ K parameters) with zero-initialized residual structure, trained with supervised InfoNCE, which is a standard global contrastive loss.
- **LoRA + Triplet** [4]: identical low-rank architecture as above, but trained with the same small-margin ( $m = 0.2$ ) triplet loss as EGA.

The two LoRA configurations isolate the effect of architectural capacity vs. loss function: comparing LoRA + InfoNCE against Icon/SRL holds the loss family roughly fixed while changing capacity, and comparing LoRA + Triplet against EGA holds the loss fixed while changing capacity.

869 **Indexing and metrics.** For all retrieval evaluations we use FAISS [30] IndexIVFFlat with  
870 nlist=10 centroids and report results at  $nprobe \in \{1, 5, 10\}$ . We measure both Label Preci-  
871 sion (LP@ $K$ , Eq. 1) and ANNS Recall (AR@ $K$ , Eq. 2) at  $K \in \{1, 3, 5, 10\}$ .

872 **Training.** All adapters are trained for 150 epochs with AdamW (weight decay  $10^{-4}$ ), cosine-  
873 annealed learning rate, and batch size 256. EGAMLP-based methods (EGA, ICon, SRL) use learning  
874 rate  $10^{-4}$ ; LoRA variants use  $10^{-3}$  to compensate for their smaller parameter count. Triplet-based  
875 methods sample one positive and one negative per anchor uniformly within each mini-batch; global  
876 contrastive methods (ICoN, SRL, LoRA + InfoNCE) operate on the full pairwise similarity matrix.  
877 All experiments use a fixed random seed (42) for reproducibility.

878 **Compute platform.** All experiments were conducted on the Chameleon Cloud [42] platform using  
879 a compute node equipped with 256 AMD EPYC-7763 CPU cores, 512 GiB of RAM, and an NVIDIA  
880 A100 GPU of 80 GB HBM2e. The operating system is Ubuntu 24.04 LTS.